HW#1; Problems Statement:

1–14 An incandescent lamp absorbs 100 W when connected to a 120-V source. A energy-efficient compact fluorescent lamp (CFL) producing the same amount of light absorbs 16 W when connected to the same source.

How much cheaper is it to operate the CFL versus the incandescent bulb over 1,000 hours when electricity costs 7.8 cents/kWh?

1–16 A string of holiday lights is protected by a 5-A fuse and has 25 bulbs, each of which is rated at 7 W. How many strings can be connected end-to-end across a 120 V circuit without blowing a fuse?

1–22 Figure P1–22 shows an electric circuit with a voltage and a current variable assigned to each of the six devices. The device voltages and currents are observed to be

	v(V)	i(A)		v(V)	i(A)
Device 1	15	−1	Device 4	−10	−1
Device 2	5	1	Device 5	20	−3
Device 3	10	2	Device 6	20	2

Find the power associated with each device and state whether the device is absorbing or delivering power. Use the power balance to check your work.

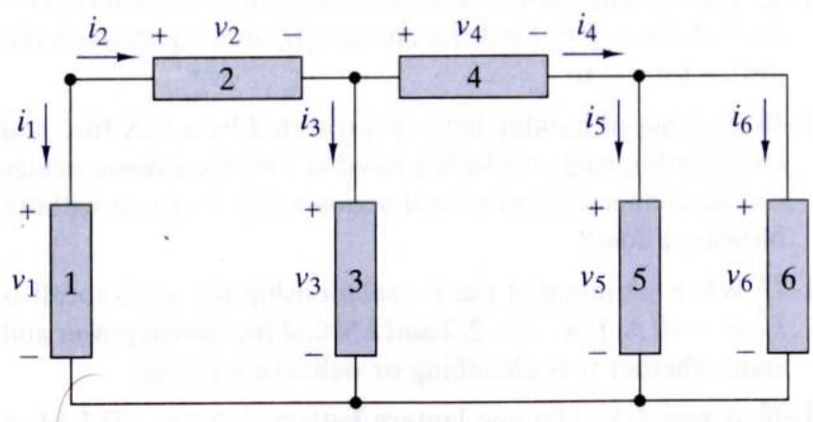

FIGURE P1–22

2–1 The current through a 56-kΩ resistor is 2.2 mA. Find the voltage across the resistor.

2–6 In Figure P2–6 find R_x and the power delivered to the resistor.

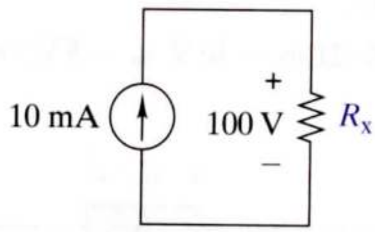

FIGURE P2–6

2–9 A 100-kΩ resistor has a power rating of 0.125 W. Find the maximum voltage that can be applied to the resistor.

2–12 A thermistor is a temperature-sensing element composed of a semiconductor material that exhibits a large change in resistance proportional to a small change in temperature. A particular thermistor has a resistance of 5 kΩ at 25 °C. Its resistance is 340 Ω at 100 °C. Assuming a straight-line relationship between these two values, at what temperature will the thermistor's resistance equal 1 kΩ?

2–15 For the circuit in Figure P2–15:

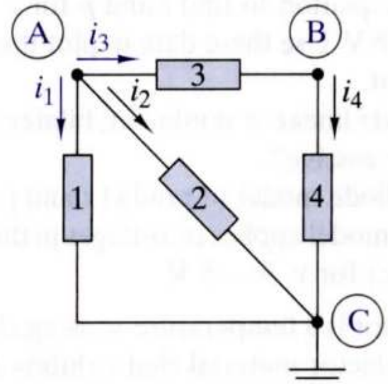

FIGURE P2–15

(a) Identify the nodes and at least two loops.
(b) Identify any elements connected in series or in parallel.
(c) Write KCL and KVL connection equations for the circuit.

2–20 The circuit in Figure P2–20 is organized around the three signal lines A, B, and C.

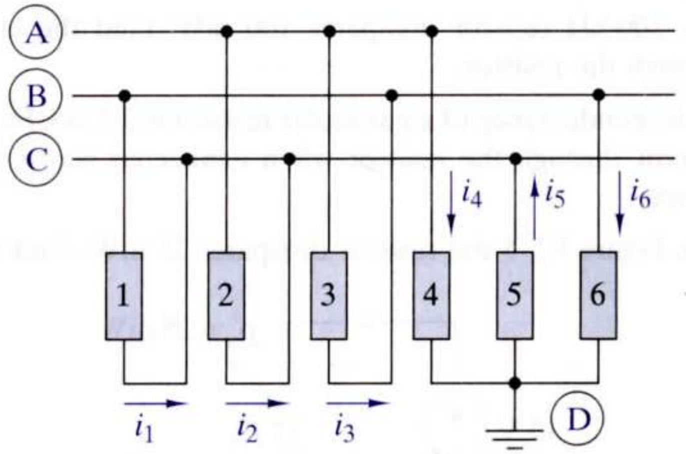

FIGURE P2–20

(a) Identify the nodes and at least three loops in the circuit.
(b) Write KCL connection equations for the circuit.
(c) If $i_1 = -20\,\text{mA}$, $i_2 = -12\,\text{mA}$, and $i_3 = 50\,\text{mA}$, find i_4, i_5, and i_6.
(d) Show that the circuit in Figure P2–20 is identical to that in Figure P2–17.

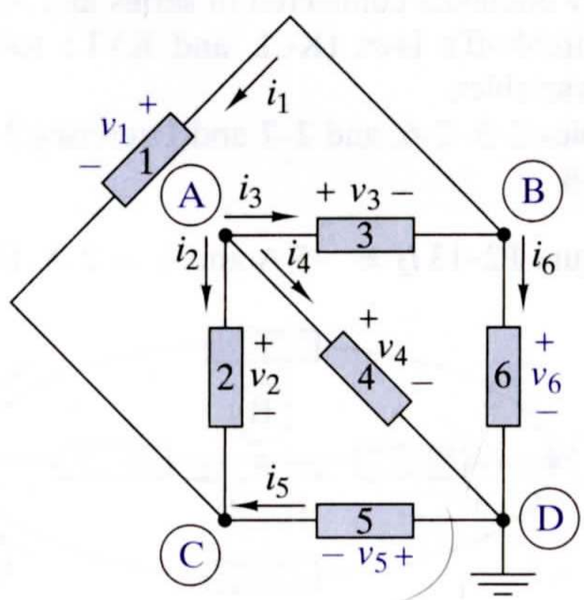

FIGURE P2–17

2–22 In Figure P2–22 $i_1 = 25$ mA, $i_2 = 10$ mA, and $i_3 = -15$ mA. Find i_4 and i_5.

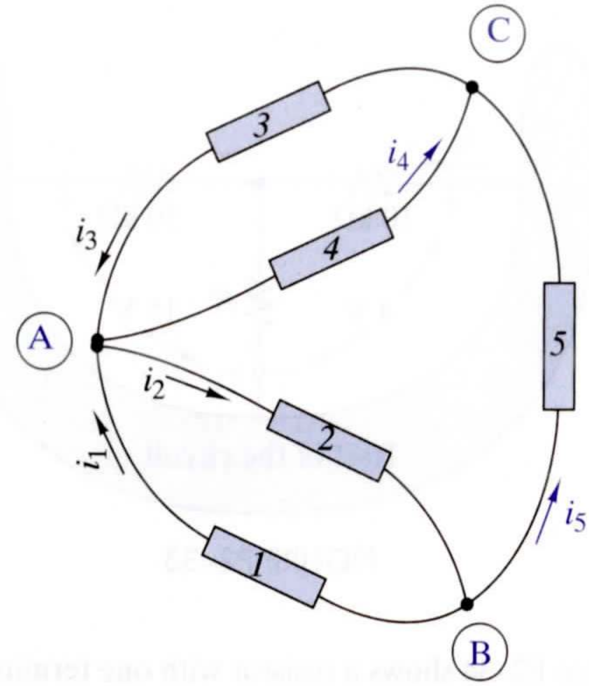

FIGURE P2–22